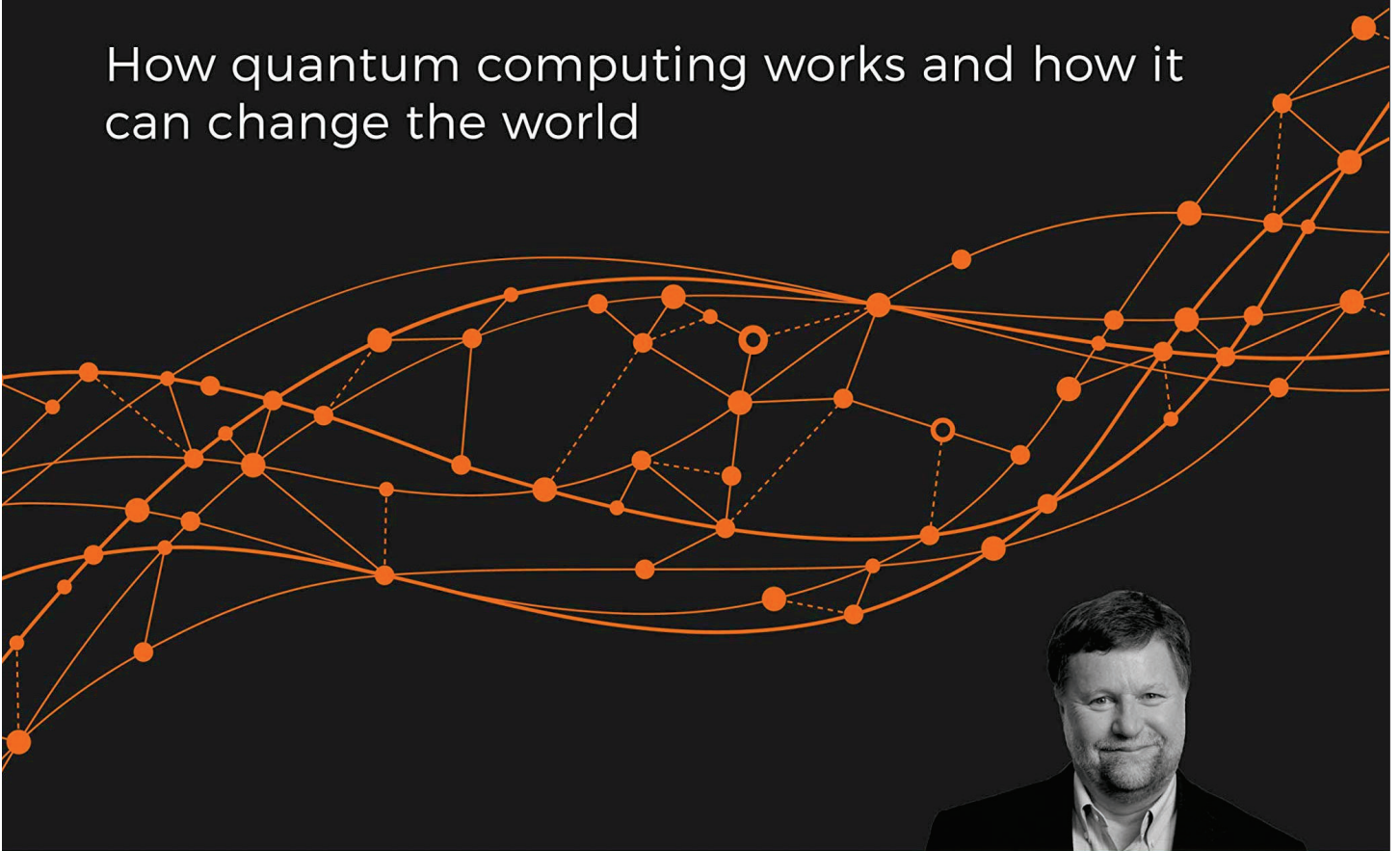


EXPERT INSIGHT

# Dancing with Qubits

How quantum computing works and how it can change the world



**Robert S. Sutor**

**Packt>**

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# Preface

*Everything we call real is made of things  
that cannot be regarded as real.*

Niels Bohr [1]

When most people think about computers, they think about laptops or maybe even the bigger machines like the servers that power the web, the Internet, and the cloud. If you look around, though, you may start seeing computers in other places. Modern cars, for example, have anywhere from around 20 computers to more than 100 to control all the systems that allow you to move, brake, monitor the air conditioning, and control the entertainment system.

The smartphone is the computer many people use more than anything else in a typical day. A modern phone has a 64-bit processor in it, whatever a “64-bit processor” is. The amount of memory used for running all those apps might be 3Gb, which means 3 gigabytes. What’s a “giga” and what is a byte?

All these computers are called *classical computers* and the original ideas for them go back to the 1940s. Sounding more scientific, we say these computers have a *von Neumann architecture*, named after the mathematician and physicist John von Neumann.

It’s not the 1940s anymore, obviously, but more than seventy years later we still have the modern versions of these machines in so many parts of our lives. Through the years, the “thinking” components, the processors, have gotten faster and faster. The amount of memory has also gotten larger so we can run more—and bigger—apps that do some pretty sophisticated things. The improvements in graphics processors have given us better and better games. The amount of storage has skyrocketed in the last couple of decades, so we can have more and more apps and games and photos and videos on devices we carry around with us. When it comes to these classical computers and the way they have developed, “more is better.”

We can say similar things about the computer servers that run businesses and the Internet around the world. Do you store your photos in the cloud? Where is that exactly? How many photos can you keep there and how much does it cost? How quickly can your photos and all the other data you need move back and forth to that nebulous place?

It's remarkable, all this computer power. It seems like every generation of computers will continue to get faster and faster and be able to do more and more for us. There's no end in sight for how powerful these small and large machines will get to entertain us, connect us to our friends and family, and solve the important problems in the world.

*Except . . . that's false.*

While there will continue to be some improvements, we will not see anything like the doubling in processor power every two years that happened starting in the mid-1960s. This doubling went by the name of *Moore's Law* and went something like "every two years processors will get twice as fast, half as large, and use half as much energy."

These proportions like "double" and "half" are approximate, but physicists and engineers really did make extraordinary progress for many years. That's why you can have a computer in a watch on your wrist that is more powerful than a system that took up an entire room forty years ago.

A key problem is the part where I said processors will get half as large. We can't keep making transistors and circuits smaller and smaller indefinitely. We'll start to get so small that we approach the atomic level. The electronics will get so crowded that when we try to tell part of a processor to do something a nearby component will also get affected.

There's another deeper and more fundamental question. Just because we created an architecture over seventy years ago and have vastly improved it, does that mean all kinds of problems can eventually be successfully tackled by computers using that design? Put another way, why do we think the kinds of computers we have now might eventually be suitable for solving every possible problem? Will "more is better" run out of steam if we keep to the same kind of computer technology? Is there something wrong or limiting about our way of computing that will prevent our making the progress we need or desire?

Depending on the kind of problem you are considering, it's reasonable to think the answer to the last question is somewhere between "probably" and "yes."

That's depressing. Well, it's only depressing if we can't come up with one or more new types of computers that have a chance of breaking through the limitations.

That's what this book is about. Quantum computing as an idea goes back to at least the early 1980s. It uses the principles of quantum mechanics to provide an entirely new kind of

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computer architecture. Quantum mechanics in turn goes back to around 1900 but especially to the 1920s when physicists started noticing that experimental results were not matching what theories predicted.

However, this is not a book about quantum mechanics. Since 2016, tens of thousands of users have been able to use quantum computing hardware via the cloud, what we call quantum cloud services. People have started programming these new computers even though the way you do it is unlike anything done on a classical computer.

Why have so many people been drawn to quantum computing? I'm sure part of it is curiosity. There's also the science fiction angle: the word "quantum" gets tossed around enough in sci-fi movies that viewers wonder if there is any substance to the idea.

Once we get past the idea that quantum computing is new and intriguing, it's good to ask "ok, but what is it really good for?" and "when and how will it make a difference in my life?" I discuss the use cases experts think are most tractable over the next few years and decades.

It's time to learn about quantum computing. It's time to stop thinking classically and to start thinking *quantumly*, though I'm pretty sure that's not really a word!

## For whom did I write this book?

This book is for anyone who has a very healthy interest in mathematics and wants to start learning about the physics, computer science, and engineering of quantum computing. I review the basic math, but things move quickly so we can dive deeply into an exposition of how to work with qubits and quantum algorithms.

While this book contains a lot of math, it is not of the definition-theorem-proof variety. I'm more interested in presenting the topics to give you insight on the relationships between the ideas than I am in giving you a strictly formal development of all results.

Another goal of mine is to prepare you to read much more advanced texts and articles on the subject, perhaps returning here to understand some core topic. You do not need to be a physicist to read this book, nor do you need to understand quantum mechanics beforehand.

At several places in the book I give some code examples using Python 3. Consider these to be extra and not required, but if you do know Python they may help in your understanding.

Many of the examples in this book come from the IBM Q quantum computing system. I was an IBM Q executive team member during the time I developed this content.

## What does this book cover?

Before we jump into understanding how quantum computing works from the ground up, we need to take a little time to see how things are done classically. In fact, this is not only for the sake of comparison. The future, I believe, will be a hybrid of classical and quantum computers.

The best way to learn about something is start with basic principles and then work your way up. That way you know how to reason about it and don't rely on rote memorization or faulty analogies.

### 1 – Why Quantum Computing?

In the first chapter we ask the most basic question that applies to this book: why quantum computing? Why do we care? In what ways will our lives change? What are the use cases to which we hope to apply quantum computing and see a significant improvement? What do we even mean by “significant improvement”?

### I – Foundations

The first full part covers the mathematics you need to understand the concepts of quantum computing. While we will ultimately be operating in very large dimensions and using complex numbers, there's a lot of insight you can gain from what happens in traditional 2D and 3D.

### 2 – They're Not Old, They're Classics

Classical computers are pervasive but relatively few people know what's inside them and how they work. To contrast them later with quantum computers, we look at the basics along with the reasons why they have problems doing some kinds of calculations. I introduce the simple notion of a bit, a single 0 or 1, but show that working with many bits can eventually give you all the software you use today.

### 3 – More Numbers than You Can Imagine

The numbers people use every day are called real numbers. Included in these are integers, rational numbers, and irrational numbers. There are other kinds of numbers, though, and structures that have many of the same algebraic properties. We look at these to lay the groundwork to understand the “compute” part of what a quantum computer does.

### 4 – Planes and Circles and Spheres, Oh My

From algebra we move to geometry and relate the two. What is a circle, really, and what does it have in common with a sphere when we move from two to three dimensions? Trigonometry becomes more obvious, though that is not a legally binding statement. What you thought of as

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a plane becomes the basis for understanding complex numbers, which are key to the definition of quantum bits, usually known as *qubits*.

## 5 – Dimensions

After laying the algebraic and geometric groundwork, we move beyond the familiar two- and three-dimensional world. Vector spaces generalize to many dimensions and are essential for understanding the exponential power that quantum computers can harness. What can you do when you are working in many dimensions and how should you think about such operations? This extra elbow room comes into play when we consider how quantum computing might augment AI.

## 6 – What Do You Mean “Probably”?

“God does not play dice with the universe,” said Albert Einstein.

This was not a religious statement but rather an expression of his lack of comfort with the idea that randomness and probability play a role in how nature operates. Well, he didn’t get that quite right. Quantum mechanics, the deep and often mysterious part of physics on which quantum computing is based, very much has probability at its core. Therefore, we cover the fundamentals of probability to aid your understanding of quantum processes and behavior.

## II – Quantum Computing

The next part is the core of how quantum computing really works. We look at quantum bits—qubits—singly and together, and then create circuits that implement algorithms. Much of this is the ideal case when we have perfect fault-tolerant qubits. When we really create quantum computers, we must deal with the physical realities of noise and the need to reduce errors.

## 7 – One Qubit

At this point we are finally able to talk about qubits in a nontrivial manner. We look at both the vector and Bloch sphere representations of the quantum states of qubits. We define superposition, which explains the common cliché about a qubit being “zero and one at the same time.”

## 8 – Two Qubits, Three

With two qubits we need more math, and so we introduce the notion of the tensor product, which allows us to explain entanglement. Entanglement, which Einstein called “spooky action at a distance,” tightly correlates two qubits so that they no longer act independently. With superposition, entanglement gives rise to the very large spaces in which quantum computations can operate.

## 9 – Wiring Up the Circuits

Given a set of qubits, how do you manipulate them to solve problems or perform calculations? The answer is you build circuits for them out of gates that correspond to reversible operations. For now, think about the classical term “circuit board.” I use the quantum analog of circuits to implement algorithms, the recipes computers use for accomplishing tasks.

## 10 – From Circuits to Algorithms

With several simple algorithms discussed and understood, we next turn to more complicated ones that fit together to give us Peter Shor’s 1995 fast integer factoring algorithm. The math is more extensive in this chapter, but we have everything we need from previous discussions.

## 11 – Getting Physical

When you build a physical qubit, it doesn’t behave exactly like the math and textbooks say it should. There are errors, and they may come from noise in the environment of the quantum system. I don’t mean someone yelling or playing loud music, I mean fluctuating temperatures, radiation, vibration, and so on. We look at several factors you must consider when you build a quantum computer, introduce Quantum Volume as a whole-system metric of the performance of your system, and conclude with a discussion of the most famous quantum feline.

This book concludes with a chapter that moves beyond today.

## 12 – Questions about the Future

If I were to say, “in ten years I think quantum computing will be able to do . . .,” I would also need to describe the three or four major scientific breakthroughs that need to happen before then. I break down the different areas in which we’re trying to innovate in the science and engineering of quantum computing and explain why. I also give you some guiding principles to distinguish hype from reality. All this is expressed in terms of motivating questions.

## References

- [1] Karen Barad. *Meeting the Universe Halfway. Quantum Physics and the Entanglement of Matter and Meaning*. 2nd ed. Duke University Press Books, 2007.

## What conventions are used in this book?

When I want to highlight something important that you should especially remember, I use this kind of box:

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This is very important.

This book does not have exercises but it does have questions. Some are answered in the text and others are left for you as thought experiments. Try to work them out as you go along. They are numbered within chapters.

**Question 0.0.1**

Why do you ask so many questions?

Code samples and output are presented to give you an idea about how to use a modern programming language, Python 3, to experiment with basic ideas in quantum computing.

```
def obligatoryFunction():  
    print("Hello quantum world!")  
  
obligatoryFunction()  
  
Hello quantum world!
```

Numbers in brackets (for example, [1]) are references to additional reading materials. They are listed at the end of each chapter in which the bracketed number appears.

**To learn more**

Here is a place where you might see a reference to learn more about some topic. [1]

## Get in touch

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Now let's get started by seeing why we should look at quantum computing systems to try to solve problems that are intractable with classical systems.